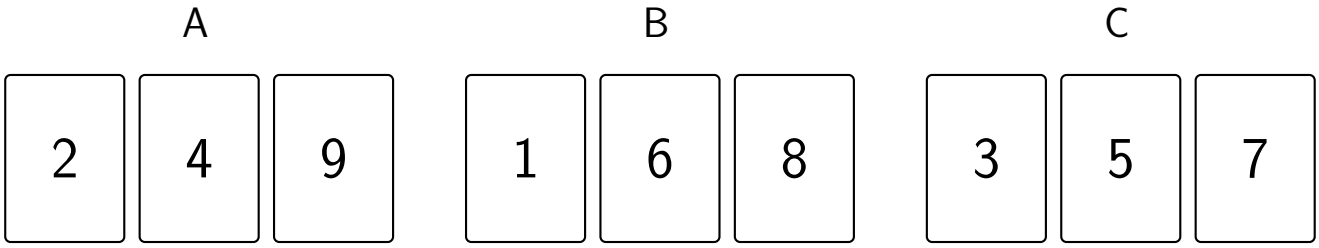
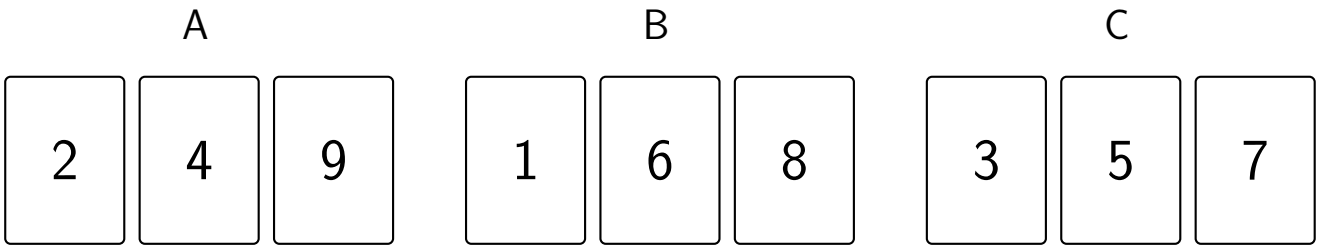
Decks A, B and C are bags of three number cards, shown below. Mix each bag on its own. In a game, each player takes one deck. Each round, draw one card from each player's bag without looking; the bigger number wins the round. Put both cards back before the next round.

Problem 1: One player picks one of the decks A, B, and C. Then the other player picks a different deck. They play ten rounds while a third player draws the cards and keeps score. Rotate jobs and play again. Is it better to pick first or second?



	First pick	Second pick
example	A, 4 wins	C, 6 wins

Problem 2: For each pair of decks, find how many of the possible card pairs each deck wins. Is there a deck that wins more than half its pairs against both others? Do the three deck totals settle this question?



Problem 3: Keep 9 in A, 8 in B, and 7 in C. Use cards 1 to 6 once each to finish the decks so A wins more than half its pairs against B, B against C, and C against A. Find every arrangement and explain why none are missing.

